

LARP-Scaler: Direct Latent Refinement for Fast Multi-Scale Image Super-Resolution

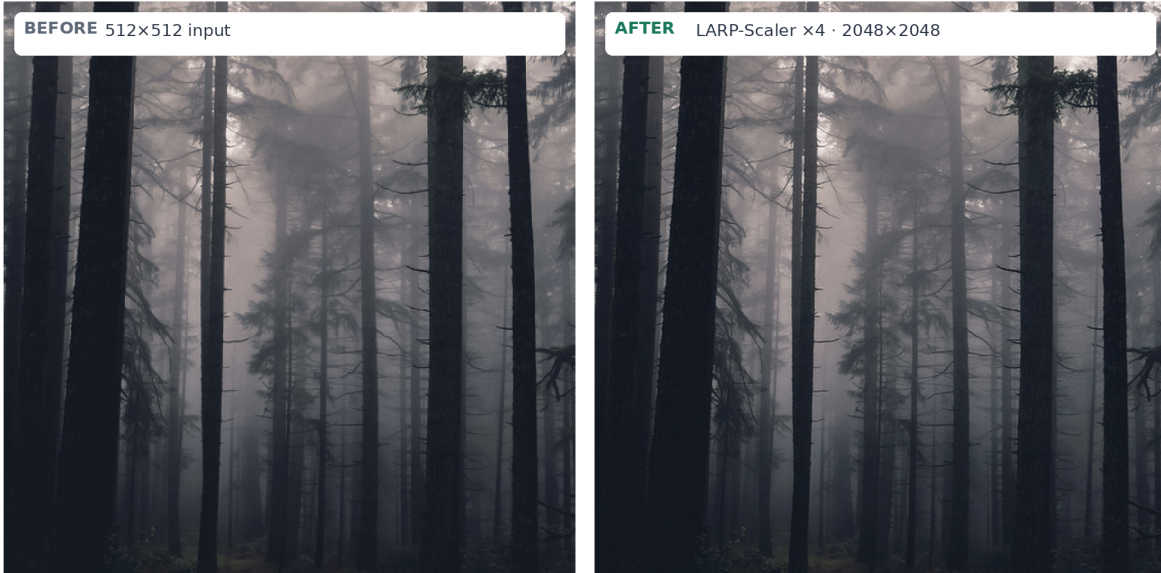
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<https://github.com/Vovanm88/LARP-Scaler>

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LARP-Scaler image upscaling

Real photo · one-pass latent refinement · $\times 4$



Abstract

Latent image generators provide expressive priors, but practical super-resolution must balance synthesized detail against exact reconstruction and runtime cost. We present LARP-Scaler, a multi-scale upscaler that starts from the VAE encoding of a conventionally resized image and directly refines this structured latent with a SANA-based linear Diffusion Transformer. A zero-initialized adapter independently encodes the low-resolution observation and injects its features through gated attention. The released runtime supports $2\times$, $4\times$, and $8\times$ enlargement, ships cached prompt tensors instead of a text encoder, and combines native VAE tiling with overlapped, shifted-grid DiT tiling.

The model is trained through a progressive curriculum on 288,423 captioned records selected by native-resolution measurements, DINOv2 features, VLM-distilled quality models, and domain-aware artifact gates. On controlled $4\times$ reconstruction, LARP-Scaler reaches 31.817 dB PSNR on 12 photographs, ahead of Real-ESRGAN (29.641 dB)

and LUA (19.554 dB), but below the protocol-matched Lanczos baseline (33.434 dB). The same distortion-generation trade-off appears on 12 anime images. A separate 48-photo, 2048-pixel matrix evaluates all three scales; at $8\times$, LARP-Scaler is within 0.086 dB PSNR of Lanczos, while interpolation retains a larger advantage at $2\times$ and $4\times$. LPIPS and DISTS results are mixed, and 95% bootstrap intervals are wide on the 12-image sets. Disabling the trained adapter branch changes PSNR by 0.514 dB, whereas matched rather than mismatched guidance accounts for 0.079 dB under the same protocol. These measurements characterize specific reconstruction settings rather than universal perceptual superiority.

1 Introduction

Single-image super-resolution seeks a plausible high-resolution image $\mathbf{x}_{\text{HR}}$ given a low-resolution observation $\mathbf{x}_{\text{LR}}$. Classical formulations emphasize fidelity to paired targets and commonly optimize pixel or perceptual reconstruction

objectives. More recent generative approaches trade exact reconstruction for an expressive image prior, improving texture synthesis under severe information loss. In practice, the useful operating point depends on the application: archival reconstruction rewards faithful pixels, while creative enlargement may prefer plausible high-frequency detail.

Latent generative models make this trade-off especially attractive. A high-resolution image can be manipulated through a spatially compressed representation, and efficient transformer backbones such as SANA [1] reduce the cost of global modelling. However, a text-to-image model is not automatically an upscaler. It must retain geometry from the input, condition on information at a different spatial scale, avoid uncontrolled regeneration, and process outputs larger than GPU memory.

LARP-Scaler addresses these requirements as a direct-refinement system. The low-resolution image is first resized to the exact target geometry with a conventional kernel and encoded by a deep-compression VAE. Rather than initializing a full generation trajectory from noise, the transformer begins from this structured latent and applies a short flow-matching schedule. The original input is encoded independently and passed through a learned multi-scale adapter. This separation supplies the transformer with both a target-sized state and a lower-resolution observation that preserves the actual image content.

The public runtime additionally treats deployment as part of the method. Text embeddings are computed once and stored with the checkpoint; the VAE and DiT remain on the accelerator in BF16; VAE and transformer tiling use independent coordinate systems; and overlapped transformer tiles are blended into a single latent canvas after every global step. These decisions are important for faithful large-image inference and prevent the coordinate drift observed when a manual VAE tiler is wrapped around the VAE’s native tiling mechanism.

Our contributions are:

- a SANA-based direct latent-refinement pipeline for $2\times$, $4\times$, and $8\times$ super-resolution;
- a zero-initialized image-guidance adapter and a text-encoder-free, tiled inference implementation; and
- a documented progressive training curriculum and 288k-image data pipeline, together with controlled quality, speed, and component ablations.

The code, checkpoint, and public demo are available at <https://github.com/Vovanm88/LARP-Scaler>, <https://huggingface.co/VladimirM388/larp scaler-v2-bf16>, and <https://huggingface.co/spaces/Anonymous/LARP-Scaler>. Human-preference evaluation and an arXiv identifier remain pending.

2 Related Work

Image super-resolution. Deep super-resolution has progressed from distortion-oriented convolutional models

to perceptual and adversarial restoration. Real-ESRGAN [2] targets blind real-world restoration using high-order synthetic degradations and remains a strong, efficient pixel-space baseline. Distortion metrics such as PSNR and SSIM [3] favour correspondence to a known target, whereas learned metrics such as LPIPS [4] and DISTS [5] better approximate perceptual similarity. We report both distortion and learned perceptual distances; human-preference evaluation remains outside the present study.

Latent generative models. Latent diffusion models [6] move denoising from pixels to a learned compressed representation. SANA [1] combines a 32-times-compressed autoencoder with a linear Diffusion Transformer to make high-resolution synthesis more efficient. LARP-Scaler reuses this latent geometry and transformer family, but changes the task from text-conditioned synthesis to image-conditioned direct refinement.

Flow matching and few-step refinement. Flow matching [7] learns a time-dependent vector field between probability paths and provides a convenient basis for continuous-time generative models. The LARP-Scaler runtime uses a FlowMatch Euler scheduler with a short sigma schedule beginning at a user-selected refinement level. The runtime permits multiple Euler steps, but the released checkpoint is evaluated with one-step direct refinement. Additional steps are not assumed to improve quality: the four-step setting was worse in internal reconstruction checks and is therefore not presented as a quality preset.

Latent and pixel decoding approaches. LUA [8] performs single-pass latent upscaling with a lightweight Swin-style adapter and scale-specific heads. PiD [9] instead reformulates decoding as conditional pixel diffusion, jointly decoding and upsampling latent representations. Both are relevant efficiency baselines but optimize different trade-offs: LUA focuses on a feed-forward latent adapter, PiD on expressive pixel-space decoding, and LARP-Scaler on iterative direct refinement of a target-sized latent.

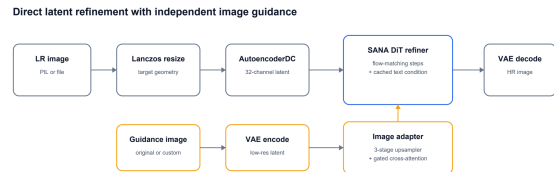


Figure 1: LARP-Scaler overview. The input is resized to the requested output geometry and encoded into the latent state refined by a SANA DiT. A separate encoding of the original or user-provided guidance image is upscaled by a three-stage adapter and injected through gated attention. Cached text tensors remove the text encoder from the runtime.

3 Method

3.1 Direct latent refinement

Let the requested enlargement factor be $s \in \{2, 4, 8\}$. We first resize the input to the padded target geometry:

$$\tilde{\mathbf{x}} = \mathcal{R}_s(\mathbf{x}_{\text{LR}}), \quad (1)$$

where $\mathcal{R}_s$ is Lanczos by default and padding makes both dimensions compatible with the VAE stride. The VAE encoder $\mathcal{E}$ produces the target-sized latent

$$\mathbf{z}_0 = \mathcal{E}(\tilde{\mathbf{x}}) \in \mathbb{R}^{B \times 32 \times H_z \times W_z}. \quad (2)$$

Unlike a noise-initialized image-generation pipeline, LARP-Scaler retains $\mathbf{z}_0$ as the initial state; no Gaussian noise is added at inference. A flow-matching transformer predicts the update field at a sequence of effective refinement levels $\{\sigma_k\}_{k=1}^K$. For the linear schedule used in the reported experiments,

$$\sigma_k = \sigma_{\max} - \frac{k-1}{K-1} \left(\sigma_{\max} - \frac{\sigma_{\max}}{K} \right), \quad (3)$$

with the one-step case defined simply as $\sigma_1 = \sigma_{\max}$. The runtime unshifts these effective values according to the checkpoint scheduler and appends the terminal value $\sigma_{K+1} = 0$. Each FlowMatch Euler update is

$$\mathbf{z}_{k+1} = \mathbf{z}_k + (\sigma_{k+1} - \sigma_k) \mathbf{v}_\theta(\mathbf{z}_k, \sigma_k, \mathbf{z}_g, s). \quad (4)$$

At inference the unknown clean component of the training interpolation path cannot be constructed, so the resized-image latent is used directly as the initial estimate. Consequently, $\sigma_{\max}$ acts as a refinement step size rather than asserting that the initial state is an exact sample from the training path.

3.2 SANA transformer backbone

The transformer extends the SANA Diffusion Transformer implementation. It keeps the linear-attention backbone, timestep-conditioned adaptive normalization, text cross-attention, and flow-matching output head, while adding an image-guidance path. The released implementation accepts 32-channel VAE latents. The scale s is supplied to the transformer so that one checkpoint can process all three enlargement factors.

Text remains useful as a weak prior even when the input image supplies most semantic information. Rather than loading a language model or text encoder at runtime, the checkpoint stores embeddings and masks for a fixed positive description and an empty prompt. When classifier-free guidance is disabled ($w \leq 1$), only the positive condition is evaluated. For $w > 1$, the unconditional and conditional predictions are combined as

$$\mathbf{v} = \mathbf{v}_u + w(\mathbf{v}_c - \mathbf{v}_u). \quad (5)$$

3.3 Image-guidance adapter

The default guidance observation is the original input, although a separate image can be supplied. It is resized to the unscaled padded geometry, encoded with the same VAE, and multiplied by a conditioning strength α :

$$\mathbf{z}_g = \alpha \mathcal{E}(\mathcal{R}_1(\mathbf{x}_g)). \quad (6)$$

The adapter maps $\mathbf{z}_g$ to target latent resolution using a 3×3 input convolution followed by up to three learned $2 \times$ upsampling stages. Each stage contains two residual convolutions, bilinear interpolation, and a post-convolution. The number of active stages is $\log_2 s$.

The resulting guidance map is projected to the transformer’s inner dimension. By default, every second transformer block performs cross-attention from the image tokens to these guidance tokens. Its output passes through a zero-initialized, timestep-conditioned gate before addition to the main residual stream. A second zero-initialized adaptive 1×1 convolution injects guidance into the first feed-forward feature map. Zero initialization lets the adapter begin as a no-op relative to the pretrained transformer and learn its contribution during fine-tuning.

3.4 Tiled inference

Large images require distinct strategies for the autoencoder and transformer. For the VAE we call the native `AutoencoderDC.enable_tiling` mechanism with a default sample tile of 1024 pixels and overlap of 256 pixels. We do not wrap this in another spatial tiler, because an outer blending coordinate system can disagree with native VAE crop geometry.

The DiT is tiled independently in latent space. Given latent tile size T and overlap O , each tile produces an updated latent $\hat{\mathbf{z}}_i$. A separable Hann window $\mathbf{W}_i$, clamped elementwise to a minimum of 10^{-3} so that outer image edges retain non-zero support, blends all predictions:

$$\mathbf{z}^{k+1} = \frac{\sum_i \mathbf{W}_i \odot \hat{\mathbf{z}}_i^{k+1}}{\sum_i \mathbf{W}_i + \epsilon}. \quad (7)$$

The accumulated weights are normalized with a minimum denominator of 10^{-6} . Every tile reads from the same globally assembled latent for a given step. On alternate steps after the first, the grid is shifted by half a stride so that boundaries do not remain fixed across the trajectory. Accumulation and weights use FP32, while VAE and transformer passes use the checkpoint dtype.

Tile batches are selected from available accelerator memory, capped at 16 tiles, and reduced by half following CUDA out-of-memory errors. Classifier-free guidance is included in this memory estimate because it doubles the model batch.

3.5 Progressive training curriculum

We did not change a text-to-image checkpoint into a direct refiner in a single optimization run. The training

curriculum progressively changes the data distribution, conditioning mechanism, and finally the flow endpoints. All stages are full-parameter updates rather than LoRA training. The curriculum is summarized in Table 1.

Stage 1: adapting the SANA prior. We first adapt all SANA DiT parameters to our curated image distribution and the Z-Image/FLUX-style flow schedule (shift 3, 1,000 training timesteps). The broad CPT phase includes a wider quality range and deliberately retained defect examples; subsequent SFT uses cleaner, more caption-aligned data. This stage contains no image-guidance adapter. Its role is to shift the image prior and flow behavior before introducing the structurally different super-resolution task.

Stages 2–3: introducing and stabilizing image guidance. We then attach the adapter described in Section 3.3. Its residual gates and first-block feed-forward injection are initialized to zero, so the new branch initially leaves the adapted backbone unchanged. The 256-channel guidance encoder, three scale-dependent upsampling stages, ten image cross-attention modules (one in every second block of the 20-block DiT), and the complete SANA backbone are optimized jointly.

For a clean latent $\mathbf{z}_{\text{HR}}$, Gaussian noise ϵ , and Bernoulli variable $b \sim \text{Bernoulli}(0.5)$, the v1 stages use

$$\mathbf{z}_{\text{src}} = \begin{cases} \mathbf{z}_{\text{deg}}, & b = 1, \\ \mathbf{z}_{\text{HR}}, & b = 0, \end{cases} \quad (8)$$

$$\mathbf{z}_{\sigma}^{\text{mixed}} = (1 - \sigma)\mathbf{z}_{\text{src}} + \sigma\epsilon, \quad (9)$$

$$\mathbf{v}_{\text{mixed}}^* = \epsilon - \mathbf{z}_{\text{HR}}. \quad (10)$$

Thus half of the examples expose the backbone to a target-sized degraded latent while the target retains the conventional clean-to-noise flow field. Independently, the adapter always receives the true low-resolution latent and the selected scale. This mixed stage teaches the new image path while preserving a generative fallback. The v1 SFT stage then resumes from v1 milestone 10,000, raises the quality threshold to 0.8, removes the retained defect subset, and retains the 0.5 mixture. We select its checkpoint 6,000 for the final stage.

Stage 4: changing the target to direct refinement. The final stage changes the problem rather than merely continuing SFT. For every example, an area-downsampled crop supplies the guidance image; random bilinear or bicubic interpolation produces $\mathbf{z}_{\text{deg}}$ at target size. Factors 2, 4, and 8 are sampled uniformly. With `refine_flow: direct` and degraded-input probability 1.0, training uses

$$\mathbf{z}_{\sigma}^{\text{direct}} = (1 - \sigma)\mathbf{z}_{\text{HR}} + \sigma\mathbf{z}_{\text{deg}}, \quad (11)$$

$$\mathbf{v}_{\text{direct}}^* = \mathbf{z}_{\text{deg}} - \mathbf{z}_{\text{HR}}. \quad (12)$$

The model therefore learns the same degraded-to-clean direction traversed by short-step inference, rather than a full noise-to-image generation path.

The predicted clean latent $\hat{\mathbf{z}}_{\text{HR}} = \mathbf{z}_{\sigma}^{\text{direct}} - \sigma\mathbf{v}_{\theta}$ is decoded at 512 pixels. The final scalar objective is

$$\mathcal{L} = \underbrace{\text{MSE}(\mathbf{v}_{\theta}, \mathbf{v}_{\text{direct}}^*)}_{\mathcal{L}_{\text{flow}}} + 0.3 \underbrace{\|\hat{\mathbf{x}}_{\text{HR}} - \mathbf{x}_{\text{HR}}\|_1}_{\mathcal{L}_{\text{recon}}} + 0.5 \underbrace{\|\Delta\hat{\mathbf{x}}_{\text{HR}} - \Delta\mathbf{x}_{\text{HR}}\|_1}_{\mathcal{L}_{\text{HF}}}, \quad (13)$$

where $\hat{\mathbf{x}}_{\text{HR}}$ is the frozen VAE-decoder output and Δ is the normalized 3×3 Laplacian response. Gradients pass through the decoder to the transformer and adapter, but decoder parameters remain frozen throughout the reported stage.

Optimization uses AdamW with learning rate 2×10^{-5} , weight decay 0.01, BF16 precision, DeepSpeed ZeRO-2, gradient clipping at 2.0, and an effective optimizer batch of 96. Training examples use 1024-pixel crops decoded from source images at up to 2048 pixels. The final-stage split reserves 1% for validation, applies 7.5% prompt dropout, and conditionally adds an image-type tag to the fixed prompt. The VAE and text encoders remain frozen; only the transformer–adapter module receives optimizer updates. Process allocation, cache production, seeds, validation cadence, and stage-boundary state are reported in Section D.

4 Training Data Pipeline

Extreme super-resolution depends on high-frequency evidence in the training targets. A nominally high-resolution image that is blurred, previously upscaled, or heavily compressed teaches the model the wrong reconstruction prior. We therefore built a metadata-first pipeline that profiles images at native resolution and preserves content diversity while filtering technical defects.

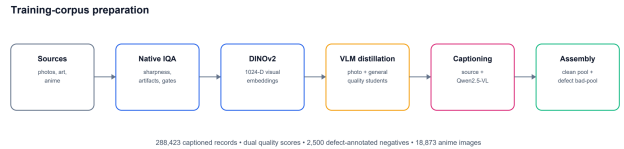


Figure 2: Data pipeline. Native-resolution measurements and DINOv2 embeddings feed two VLM-distilled student quality models. Deterministic gates, captions, and a deliberately retained defect pool produce a training-ready corpus without copying the source images.

4.1 Sources and native-resolution profiling

The final metadata index contains 288,423 records from CommonCatalog, PD12M, 4KLSDB, Unsplash, DIV8K, a small curated and synthetic fine-t2i set, and three high-resolution SFW anime sources (yande, konachan, and danbooru). The pipeline stores source references and derived metadata rather than duplicating the image corpus.

Table 1: Progressive curriculum leading to the released checkpoint. “Mixed” denotes the v1 objective that retains the original noise endpoint while replacing the clean-side input by a degraded latent with probability 0.5.

Stage	Initialization and trainable modules	Path / objective	Purpose
Domain and schedule adaptation	Upstream SANA; all DiT parameters	Standard clean-noise flow matching on our curated images under the Z-Image schedule	Move the generative prior toward the target image distribution before asking it to restore LR observations.
Adapter warm-up (v1 CPT)	Adapted SANA plus a new zero-initialized image branch; backbone and adapter jointly trainable	Mixed clean/degraded source with probability 0.5; conventional flow target	Learn multi-scale LR conditioning without abruptly discarding the generative behavior.
Adapter SFT	v1 milestone 10,000; all transformer and adapter parameters	Same mixed path on a stricter quality-filtered subset without the retained defect subset	Consolidate restoration and conditioning on cleaner targets; checkpoint 6,000 initializes the released final stage.
Direct refinement	v1-SFT checkpoint 6,000; all transformers and adapter parameters	100% degraded-to-clean latent path plus pixel and high-frequency losses	Match the short refinement trajectory used by the public runtime.

Each image is decoded at native resolution. Measurements include JPEG block-edge energy, a downsample-upsample Laplacian ratio, flat-patch noise, local and global Laplacian statistics, Tenengrad, spectral slope, high-frequency energy, edge density, JPEG quantization estimates, bytes per pixel, borders, saturation, grayscale status, and a perceptual hash. The down-up ratio was the most useful blur/prior-upscale detector and achieved approximately 0.992 AUC on a labelled validation set. In a diagnostic CommonCatalog sample, 72% of images triggered at least one native-resolution super-resolution flag that had been missed by the earlier thumbnail test.

4.2 VLM-distilled scoring

We extract the CLS embedding from DINOv2-large [10] after a 224-pixel resize. A photo-quality VLM rubric is informative for photographic content but systematically penalizes illustration. We therefore collect two teacher label sets: approximately 21k photo-rubric labels and 7.3k content-agnostic labels under a fixed annotation budget.

Two small multi-layer perceptrons consume the DINOv2 embedding concatenated with standardized pixel features. Each has a scalar quality head and a watermark/overlay head. The photo student reaches Spearman $\rho \approx 0.60$ against its teacher; the content-agnostic student reaches $\rho \approx 0.36$. Although imperfect as absolute quality estimators, the two scores are inexpensive to evaluate over the full corpus and expose different domain biases.

4.3 Gates, captions, and assembly

Three universal gates flag extreme blockiness, softness, and noise. Additional photo-oriented gates for local sharpness, borders, over-compression, and low edge density are stored but disabled by default because they over-fire on

flat-shaded art. The default base-corpus decision combines universal gates, overlay probability, and a general quality threshold of 60 on a 0–100 scale.

Existing source captions are reused where available. Missing short and long captions are generated by a local Qwen2.5-VL model [11]. Rather than discarding every low-scoring item, we uniformly sample 2,500 examples across the rejected score range. A VLM assigns one or more of 18 defect tags, and their captions explicitly state the defect. This retained defect subset provides quality-conditioned negative examples.

Anime reveals a useful domain-shift case. The content-agnostic student model rates the images reasonably, while photo-calibrated softness and low-detail gates reject many valid flat-shaded illustrations. For the anime expansion we therefore keep images based on the learned score and caption presence, without the universal pixel gate. This yields 18,873 SFW anime records. Exact source counts are reported in Table 9.

5 Experiments

5.1 Evaluation protocols

Reference reconstruction. For each headline quality set, a known 512-pixel target is downsampled to 128 pixels with Lanczos and upsampled by a factor of four. We report RGB, full-frame PSNR, SSIM, and MAE on a $[0, 1]$ range; SSIM uses an 11×11 uniform window. The photo and anime sets each contain 12 public or externally curated raster images. Bicubic and Lanczos resize the saved low-resolution inputs directly to the target size and therefore act as deterministic, protocol-matched interpolation baselines. All methods in a shared table receive the same inputs.

LARP-Scaler uses the evaluated reconstruction preset:

one refinement step, $\sigma_{\max} = 0.35$, guidance scale 1, BF16, and VAE tiling. The same settings are used for the ablation below, so Tables 2 and 10 differ only by runtime caching details. We compute 95% percentile bootstrap confidence intervals over images with 10,000 resamples. The intervals are reported in Section C; per-image values are provided in `paper/data/revision/controlled_x4_12_per_image.csv` and `paper/data/revision/revision_on_benchmarks.json`.

Expanded multi-scale reconstruction. A separate matrix uses 48 external photographs with 2048-pixel center-crop targets and Lanczos-downsampled inputs at $2\times$, $4\times$, and $8\times$. It uses the same one-step LARP-Scaler preset and the same interpolation and metric implementations. This matrix evaluates scale behavior but is not pooled with the two 12-image headline sets.

Latency. All timing uses an NVIDIA GeForce RTX 5090 with models already resident in VRAM. Each cell includes 24 images, three warm-up calls per image and ten timed calls, for 240 calls. The interval covers PIL file open, preprocessing, VAE encoding when applicable, inference, and final image conversion. Model loading and output-file writing are excluded. Native JPEG sizes can be much larger than the 1024-pixel output, so the values are end-to-end file-processing latency. Real-ESRGAN is natively $4\times$; its output is resized for requested $2\times$ and $8\times$ conditions.

Ablation. Prompt and image-adapter components are evaluated on 12 photographs at 128-to-512 $4\times$ reconstruction with one refinement step, $\sigma_{\max} = 0.35$, guidance scale 1, and BF16. Latent caches are disabled and every configuration receives the same images. Exact deterministic settings are listed in Section D.

5.2 Training trace

The final direct-refinement run lasted 49,006 s (13 h 36 min 46 s). Peak allocated GPU memory was 21.77 GB. Validation loss decreased from 0.1954 at update 100 to 0.1452 at update 4,000, a 25.69% relative reduction. Machine-readable provenance and milestone values are included in `paper/data/training_run.json`.

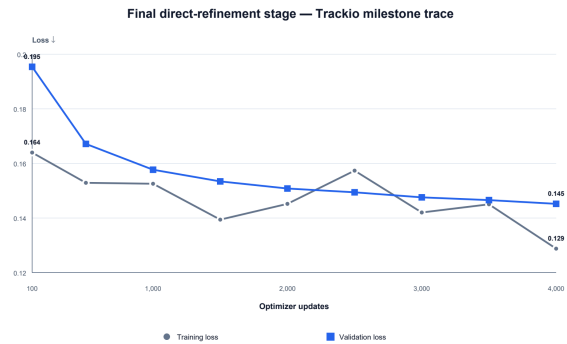


Figure 3: Trackio milestones from the final 4,000-update direct-refinement stage. Training loss is a stochastic per-update measurement; validation is evaluated every 100 updates, with selected milestones shown here.

5.3 Controlled reconstruction

Table 2: Shared $4\times$ photo reconstruction set (12 images). Interpolation baselines are evaluated on the saved Lanczos-downsampled inputs.

Method	PSNR $\uparrow$	SSIM $\uparrow$	MAE $\downarrow$
Lanczos	33.4340	0.8308	0.01910
Bicubic	33.2358	0.8248	0.01943
LARP-Scaler	31.8170	0.7877	0.02314
Real-ESRGAN	29.6409	0.7587	0.02846
LUA	19.5536	0.4071	0.09247

Table 3: Shared $4\times$ anime reconstruction set (12 images). Interpolation baselines are evaluated on the saved Lanczos-downsampled inputs.

Method	PSNR $\uparrow$	SSIM $\uparrow$	MAE $\downarrow$
Lanczos	28.7939	0.8650	0.02184
Bicubic	28.3830	0.8614	0.02190
LARP-Scaler	28.4394	0.8560	0.02382
Real-ESRGAN	26.7850	0.8695	0.02559
LUA	17.2268	0.3653	0.10970

Tables 2 and 3 separate two comparisons. Among the learned methods, LARP-Scaler has the highest PSNR and lowest MAE on both sets, including a 2.176 dB photo PSNR margin over Real-ESRGAN. The interpolation baselines, however, retain more reference pixels: Lanczos leads photo PSNR by 1.617 dB and anime PSNR by 0.355 dB over LARP-Scaler. This is expected because the inputs themselves are produced by Lanczos and interpolation does not synthesize new texture that can be penalized by paired distortion metrics. The 95% bootstrap intervals are wide and overlapping on these small sets. Real-ESRGAN retains the highest anime SSIM, further illustrating that no single metric defines the ranking.

Table 4: Separate multi-scale reconstruction matrix on 48 external photographs with 2048-pixel targets. All rows use the same target crops and Lanczos-downsampled inputs.

Scale	Method	PSNR $\uparrow$	SSIM $\uparrow$	MAE $\downarrow$
2 $\times$	Lanczos	36.6473	0.9120	0.01287
	Bicubic	36.1985	0.9059	0.01336
	LARP-Scaler	30.2797	0.7641	0.02451
4 $\times$	Lanczos	32.0484	0.8221	0.01973
	Bicubic	31.8412	0.8168	0.02000
	LARP-Scaler	30.7892	0.7875	0.02272
8 $\times$	Lanczos	29.2018	0.7348	0.02532
	Bicubic	29.0510	0.7303	0.02556
	LARP-Scaler	29.1155	0.7336	0.02589

At 8 $\times$, the LARP-Scaler PSNR gap to Lanczos narrows to 0.086 dB and LARP-Scaler exceeds Bicubic by 0.064 dB; their confidence intervals overlap. At 2 $\times$ and 4 $\times$, conservative interpolation retains a clear distortion advantage. The matrix therefore supports multi-scale operation, but not distortion-metric leadership over interpolation.

PiD is designed to synthesize high-resolution pixel detail from latent conditioning. We therefore exclude it from the 512-pixel headline tables: the available public PiD_res2k_sr4x checkpoint targets roughly 2048-pixel outputs, and its archived 512-pixel results exhibit a regular lattice artifact. The native-resolution experiment is reported separately below. LUA optimizes a different point again: a single feed-forward latent adapter with scale-specific heads, which is inexpensive but is not trained to restore the exact reference pixels of this protocol. The reported reconstruction gaps for both methods should therefore not be interpreted as a general aesthetic ranking; they quantify agreement with a known target under one protocol.

Table 5: Native-resolution PiD 4 $\times$ photo reconstruction (12 real images). Each ground-truth target is a 2048-pixel crop from an original 3–7K photograph and the input is its 512-pixel Lanczos downsample.

Method	PSNR $\uparrow$	SSIM $\uparrow$	MAE $\downarrow$
LARP-Scaler	31.4801	0.8114	0.02061
Real-ESRGAN	29.2505	0.7786	0.02571
PiD (native 2k)	26.1275	0.6683	0.03635
LUA	Flux-VAE OOM at 2048 px (32 GB)		

At its intended 2048-pixel output, PiD no longer exhibits the regular lattice artifact. Its reconstruction scores remain below LARP-Scaler and Real-ESRGAN on this strict paired protocol, but they are valid measurements of the released PiD checkpoint rather than an out-of-regime failure.

5.4 Perceptual metrics

Table 6: Perceptual distances on the shared 512-pixel 4 $\times$ sets (12 images per domain; lower is better). LPIPS uses AlexNet and DISTS uses piq. PiD is omitted here because its archived 512-pixel outputs are out of regime; LUA per-image outputs were not archived.

Domain	Method	LPIPS $\downarrow$	DISTS $\downarrow$
Photo	LARP-Scaler	0.4221	0.1766
	Real-ESRGAN	0.3778	0.1950
Anime	LARP-Scaler	0.2624	0.1215
	Real-ESRGAN	0.1537	0.1215

Table 6 complements distortion metrics. LARP-Scaler leads on DISTS for photographs and matches Real-ESRGAN on anime, while Real-ESRGAN has lower LPIPS on both sets. A distortion-oriented lead thus does not transfer uniformly to learned perceptual distances.

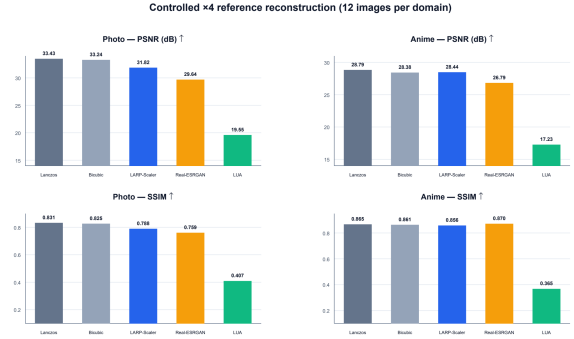


Figure 4: PSNR and SSIM on the controlled photo and anime reconstruction sets. Exact values and MAE are reported in Tables 2 and 3.

5.5 Latency

Table 7: Median end-to-end latency in seconds for 1024-pixel output on 24 photographs. Each cell aggregates 240 calls. Lower is better.

Method	2 $\times$	4 $\times$	8 $\times$
LARP-Scaler	0.8119	0.8227	0.8047
LUA	2.6323	0.6291	0.6668
Real-ESRGAN	1.0253	0.6681	0.6134

The 1024-pixel PiD latency row is excluded because it uses the same 2k-native checkpoint out of its intended output regime. At 4 $\times$, LARP-Scaler is 30.8% slower than LUA and 23.1% slower than Real-ESRGAN. At 2 $\times$ it has the lowest median in this protocol, while Real-ESRGAN is fastest at 8 $\times$. Consequently, the evidence supports a competitive quality–latency trade-off, not universal speed leadership.

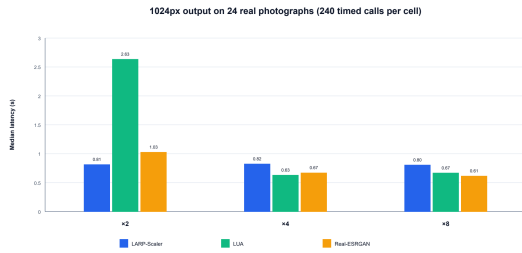


Figure 5: Median end-to-end file-processing latency. The large LUA 2 $\times$ value illustrates why performance should be reported per scale rather than collapsed to a single average.

5.6 Prompt and adapter ablation

Table 8: Selected ablation changes on 12 photographs. MSE reduction is computed from ΔPSNR as $1 - 10^{-\Delta/10}$.

Change	ΔPSNR	MSE red.
Detailed prompt, adapter off	+0.0166	0.38%
Detailed prompt, adapter on	+0.0057	0.13%
Adapter branch, tagged prompt	+0.5138	11.16%
Matched vs. mismatched guidance	+0.0788	1.80%

Measured on the same 12 photographs and the same one-step protocol, the released checkpoint improves on the untrained SANA backbone by 10.555 dB (21.2628 to 31.8178 dB in the tagged-prompt, matched-guidance configuration; see Table 10). That gap is the combined effect of the direct-refinement training stage and the guidance branch, and unlike the cross-protocol comparison in Section E it is measured on identical inputs.

Enabling the trained adapter branch is the largest measured architectural change within the released system: in the tagged-prompt configuration it raises PSNR from 31.3040 to 31.8178 dB. Replacing mismatched guidance by the matched low-resolution image accounts for only 0.0788 dB. This suggests a measurable but small content-specific contribution under this protocol; the larger branch-on difference should not be attributed entirely to matched image content. Prompt gains are positive but small and shrink after the adapter is enabled.

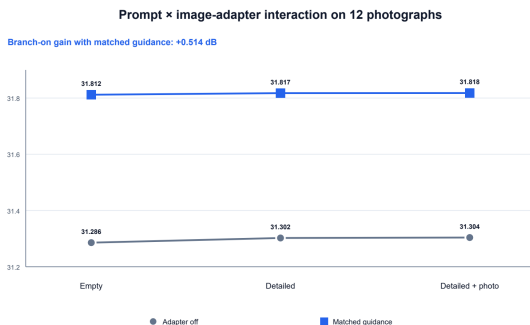


Figure 6: Prompt–adapter interaction. The y-axis is intentionally narrow to expose the prompt effect; the gap between curves is much larger.

5.7 Qualitative comparison

Figure 7 shows the released checkpoint against Real-ESRGAN and PiD on four photographs from the native-resolution benchmark. Every method receives the same 512×512 LR input and produces a 2048×2048 output. PiD therefore runs at the intended resolution of its released `res2k_sr4x` checkpoint, with four distilled steps, rather than being forced into the invalid 512-pixel-output regime. The displayed files are the same outputs scored in Table 5. Real-ESRGAN tends toward smooth, conservative texture; PiD can introduce plausible high-frequency detail; and LARP-Scaler remains closer to the paired reference on these examples.

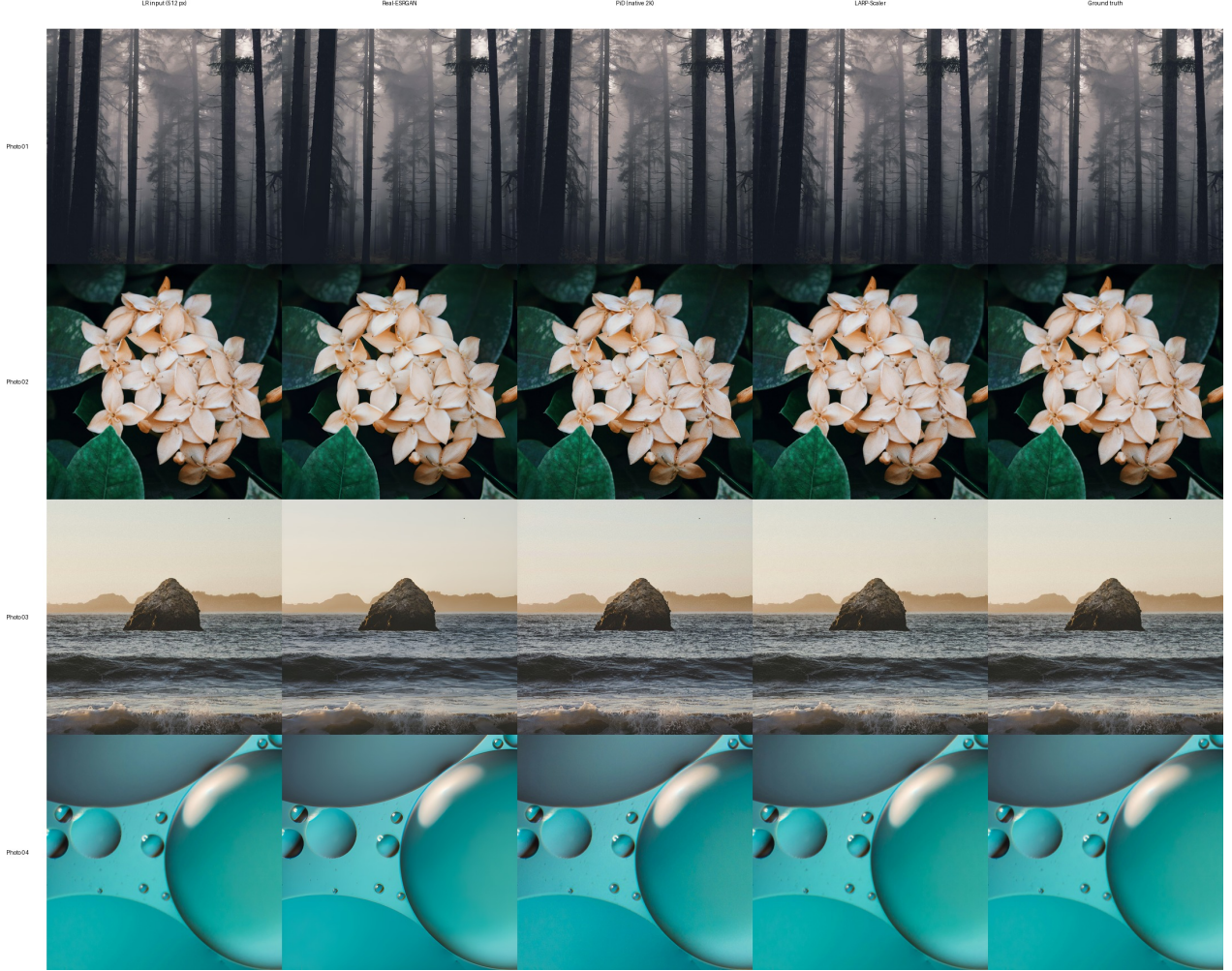


Figure 7: Native-2K qualitative comparison at $4\times$ ($512\rightarrow 2048$). PiD uses its official `PiD_res2k_sr4x` four-step operating regime; all columns share the same LR inputs and targets. The four illustrative scenes cover fog and vertical edges, flower structure, water and rock detail, and smooth specular surfaces. Aggregate claims use all twelve images in Table 5; these four rows are not a separate benchmark. The LR column is enlarged with nearest-neighbour sampling to expose its actual pixels.

6 Limitations

Small controlled sets. The main quality comparisons contain 12 photographs and 12 anime images. They are controlled, reproducible reconstruction tests, but are not broad enough to characterize rare domains, difficult degradations, faces, text, or pathological compression. The separate 48-photo matrix improves scale coverage but does not replace a broad public benchmark; bootstrap intervals on the 12-image sets are wide.

Metric coverage. PSNR, SSIM, and MAE emphasize exact reconstruction and do not fully represent perceptual quality. We additionally report LPIPS and DISTS (Table 6), which show that a distortion-oriented lead does not transfer uniformly to learned perceptual distances. No-reference IQA, FID, and human preference remain unmeasured for the released checkpoint.

Protocol separation. Several additional measurements exist at other resolutions and on other image sets. We report them separately in the appendix and do not combine them with the shared tables. Cross-protocol deltas can suggest robustness but cannot be added or treated as paired comparisons.

Hardware. Timing results use a single GPU family, the RTX 5090. Relative ordering may change with memory bandwidth, kernel support, precision, CPU-side JPEG decode, or different batch policies.

Compute availability. Access to computational resources was limited. Model development, the final training stage, ablations, and evaluation used one four-GPU RTX 5090 machine. This limited broad hyperparameter searches, multi-seed training, and evaluation at substantially larger scale.

Training lineage and release artifacts. The recovered config and Trackio trace fully identify the final direct-refinement stage, but the released model starts from a predecessor SFT checkpoint at update 6,000. Consequently, 13.6 hours is final-stage wall time rather than total end-to-end pretraining compute. The predecessor run is identified in the machine-readable provenance, but its earlier initialization contains restarted counters and is not summarized as a single total-compute claim. The arXiv identifier remains pending.

7 Conclusion

We introduced LARP-Scaler, a SANA-based system that directly refines a target-sized latent while conditioning on an independently encoded low-resolution image. Cached text tensors and two-level tiling make the same checkpoint usable at $2\times$, $4\times$, and $8\times$. On the two 12-image controlled sets, LARP-Scaler leads the tested learned baselines in PSNR and MAE, while protocol-matched interpolation retains the overall distortion advantage. The separate 48-photo matrix confirms operation at all three scales and shows that the PSNR gap to Lanczos narrows to 0.086 dB at $8\times$. Learned perceptual metrics do not produce a uniform ranking. The adapter-on ablation measures a 0.514 dB branch effect, but matched rather than mismatched guidance explains only 0.079 dB under the same protocol. These results position LARP-Scaler as a practical generative refinement system rather than a replacement for conservative interpolation on paired distortion metrics. Broader public benchmarks, real degradations, and human preference are the main remaining evaluation priorities.

References

- [1] Enze Xie, Junsong Chen, Junyu Chen, Han Cai, Haotian Tang, Yujun Lin, Zhekai Zhang, Muyang Li, Ligeng Zhu, Yao Lu, and Song Han. SANA: Efficient high-resolution image synthesis with linear diffusion transformers. In *International Conference on Learning Representations*, 2025.
- [2] Xintao Wang, Liangbin Xie, Chao Dong, and Ying Shan. Real-ESRGAN: Training real-world blind super-resolution with pure synthetic data. In *IEEE/CVF International Conference on Computer Vision Workshops*, pages 1905–1914, 2021.
- [3] Zhou Wang, Alan C. Bovik, Hamid R. Sheikh, and Eero P. Simoncelli. Image quality assessment: From error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4):600–612, 2004.
- [4] Richard Zhang, Phillip Isola, Alexei A. Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 586–595, 2018.
- [5] Keyan Ding, Kede Ma, Shiqi Wang, and Eero P. Simoncelli. Image quality assessment: Unifying structure and texture similarity. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(5): 2567–2581, 2022.
- [6] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10684–10695, 2022.
- [7] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling. In *International Conference on Learning Representations*, 2023.
- [8] Aleksandr Razin, Danil Kazantsev, and Ilya Makarov. One small step in latent, one giant leap for pixels: Fast latent upscale adapter for your diffusion models. *arXiv preprint arXiv:2511.10629*, 2025.
- [9] Yifan Lu, Qi Wu, Jay Zhangjie Wu, Zian Wang, Huan Ling, Sanja Fidler, and Xuanchi Ren. PiD: Fast and high-resolution latent decoding with pixel diffusion. *arXiv preprint arXiv:2605.23902*, 2026.
- [10] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaa El-Nouby, Mahmoud Assran, Nicolas Ballas, Wojciech Galuba, Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat, Vasu Sharma, Gabriel Synnaeve, Hu Xu, Hervé Jégou, Julien Mairal, Patrick Labatut, Armand Joulin, and Piotr Bojanowski. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024.
- [11] Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibao Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yanzhi Zhu, Mingkun Yang, Zhaohai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, Jiabo Ye, Xi Zhang, Tianbao Xie, Zesen Cheng, Hang Zhang, Zhibo Yang, Haiyang Xu, and Junyang Lin. Qwen2.5-VL technical report. *arXiv preprint arXiv:2502.13923*, 2025.

A Dataset Composition

Table 9: Final training metadata by source.

Source	Records
CommonCatalog	100,046
PD12M	99,558
4KLSDB	43,977
Unsplash	24,849
yande	8,979
konachan	6,912
danbooru	2,982
DIV8K	1,000
fine-t2i	120
Total	288,423

Training corpus composition — 288,423 records

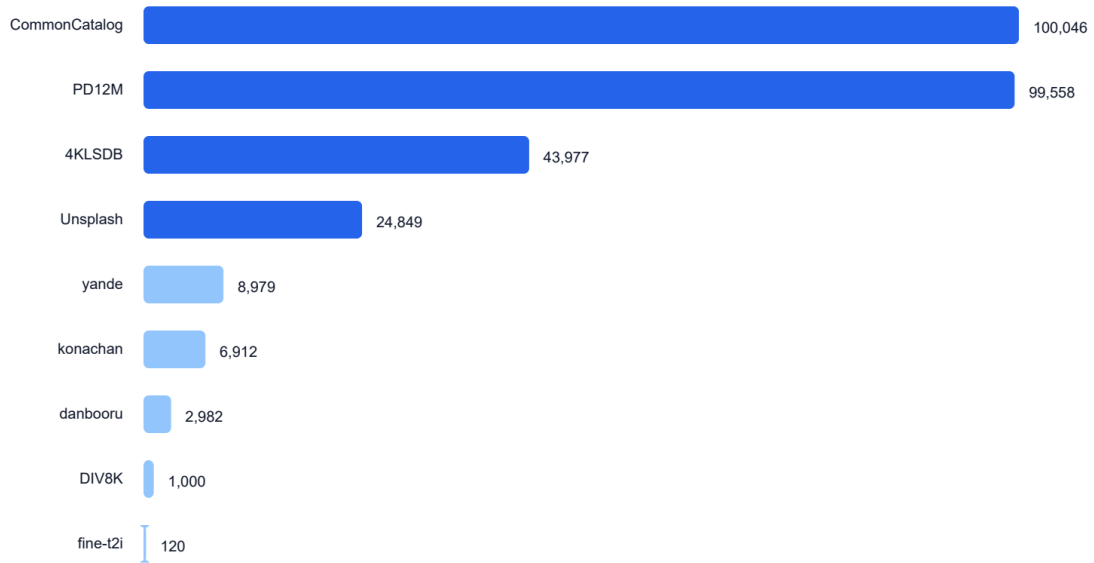


Figure 8: Dataset source counts on a linear scale.

B Full Prompt–Adapter Matrix

Table 10: Prompt and adapter states, sorted by component rather than score.

Prompt	Adapter	PSNR $\uparrow$	SSIM $\uparrow$	MAE $\downarrow$
<i>Untrained SANA backbone (no direct-refinement stage)</i>				
Empty	Off	21.258000	0.696710	0.075979
Detailed	Off	21.255834	0.696634	0.076013
Detailed + photo	Off	21.262759	0.696799	0.075949
<i>Released checkpoint</i>				
Empty	Off	31.285856	0.783891	0.024537
Detailed	Off	31.302412	0.783989	0.024495
Detailed + photo	Off	31.303990	0.783969	0.024491
Empty	Matched guidance	31.811783	0.787725	0.023156
Detailed	Matched guidance	31.817450	0.787747	0.023143
Detailed + photo	Matched guidance	31.817791	0.787734	0.023143
Detailed + photo	Mismatched guidance	31.738986	0.787650	0.023208

C Bootstrap Confidence Intervals

We use percentile bootstrap intervals over images with 10,000 resamples and seed 1234. Intervals are computed independently for each method, domain, scale, and metric; no results from the 12-image and 48-image protocols are pooled.

Table 11: Controlled 4 $\times$ reconstruction means and 95% bootstrap confidence intervals. Each domain contains 12 images.

Domain	Method	PSNR [95% CI]		SSIM [95% CI]		MAE [95% CI]	
Photo	Lanczos	33.434	[29.756, 37.220]	0.831	[0.774, 0.888]	0.0191	[0.0123, 0.0270]
	Bicubic	33.236	[29.538, 37.017]	0.825	[0.767, 0.883]	0.0194	[0.0124, 0.0275]
	LARP-Scaler	31.817	[28.302, 35.267]	0.788	[0.727, 0.851]	0.0231	[0.0149, 0.0326]
	Real-ESRGAN	29.641	[26.485, 32.701]	0.759	[0.688, 0.832]	0.0285	[0.0184, 0.0397]
	LUA	19.554	[17.570, 21.441]	0.407	[0.313, 0.502]	0.0925	[0.0728, 0.1141]
Anime	Lanczos	28.794	[25.978, 31.982]	0.865	[0.809, 0.912]	0.0218	[0.0155, 0.0291]
	Bicubic	28.383	[25.739, 31.304]	0.861	[0.806, 0.908]	0.0219	[0.0157, 0.0291]
	LARP-Scaler	28.439	[25.931, 31.187]	0.856	[0.802, 0.901]	0.0238	[0.0177, 0.0308]
	Real-ESRGAN	26.785	[25.434, 28.047]	0.869	[0.828, 0.901]	0.0256	[0.0216, 0.0305]
	LUA	17.227	[16.441, 18.025]	0.365	[0.329, 0.402]	0.1097	[0.1004, 0.1190]

Table 12: Separate 48-photo multi-scale reconstruction means and 95% bootstrap confidence intervals.

Scale	Method	PSNR [95% CI]		SSIM [95% CI]		MAE [95% CI]	
2 $\times$	Lanczos	36.647	[34.739, 38.495]	0.912	[0.884, 0.936]	0.0129	[0.0095, 0.0171]
	Bicubic	36.199	[34.314, 38.031]	0.906	[0.876, 0.931]	0.0134	[0.0099, 0.0177]
	LARP-Scaler	30.280	[28.637, 31.861]	0.764	[0.716, 0.808]	0.0245	[0.0194, 0.0306]
4 $\times$	Lanczos	32.048	[30.322, 33.710]	0.822	[0.777, 0.862]	0.0197	[0.0151, 0.0253]
	Bicubic	31.841	[30.127, 33.491]	0.817	[0.770, 0.858]	0.0200	[0.0154, 0.0256]
	LARP-Scaler	30.789	[29.149, 32.363]	0.787	[0.741, 0.829]	0.0227	[0.0179, 0.0284]
8 $\times$	Lanczos	29.202	[27.625, 30.714]	0.735	[0.681, 0.785]	0.0253	[0.0200, 0.0316]
	Bicubic	29.051	[27.486, 30.550]	0.730	[0.676, 0.781]	0.0256	[0.0202, 0.0318]
	LARP-Scaler	29.115	[27.566, 30.602]	0.734	[0.681, 0.782]	0.0259	[0.0206, 0.0320]

D Implementation and Deterministic Settings

The final direct-refinement stage uses training seed 6769, static loss scale 64, 1% warm-up and cooldown, and a plateau controller with factor 0.8 and patience 4. Three training workers use batch size 2 with 16 accumulation steps, giving effective batch size 96; a fourth GPU produces frozen VAE and text caches. The validation split uses seed 52. Validation runs every 100 updates for eight batches with seed 1234. Stage boundaries restore BF16 model weights but restart optimizer moments, scheduler history, RNG state, and data order.

All reported LARP-Scaler reconstruction and ablation runs use inference seed 1234. The bootstrap uses an independent NumPy generator with seed 1234. Full worker-level configuration and provenance remain machine-readable in `paper/data/training_run.json`.

E Additional Cross-Protocol Measurements

The following results provide context but are not merged with the main shared tables:

- On 48 external photographs with 2048-pixel targets, the full trained system improves over native SANA by 8.976 dB at the best reported $4\times$ setting.
- On a separate 32-photo adapter-control test, matched guidance exceeds mismatched guidance by 1.056 dB. This is larger than the 0.079 dB content effect on the 512-pixel ablation, demonstrating strong protocol dependence.
- Direct-refine checkpoint 4000 exceeds checkpoint 1000 by 0.047 dB on the reported 12-photo test.

These values should be read as robustness observations, not pooled estimates.

F Final-Stage Training Milestones

Table 13: Selected Trackio milestones for the released direct-refinement stage. The scalar training loss includes flow, reconstruction, and high-frequency components.

Update	Train loss	Validation loss
100	0.1640	0.1954
500	0.1529	0.1672
1,000	0.1526	0.1577
1,500	0.1394	0.1534
2,000	0.1452	0.1508
2,500	0.1574	0.1494
3,000	0.1420	0.1476
3,500	0.1450	0.1466
4,000	0.1288	0.1452